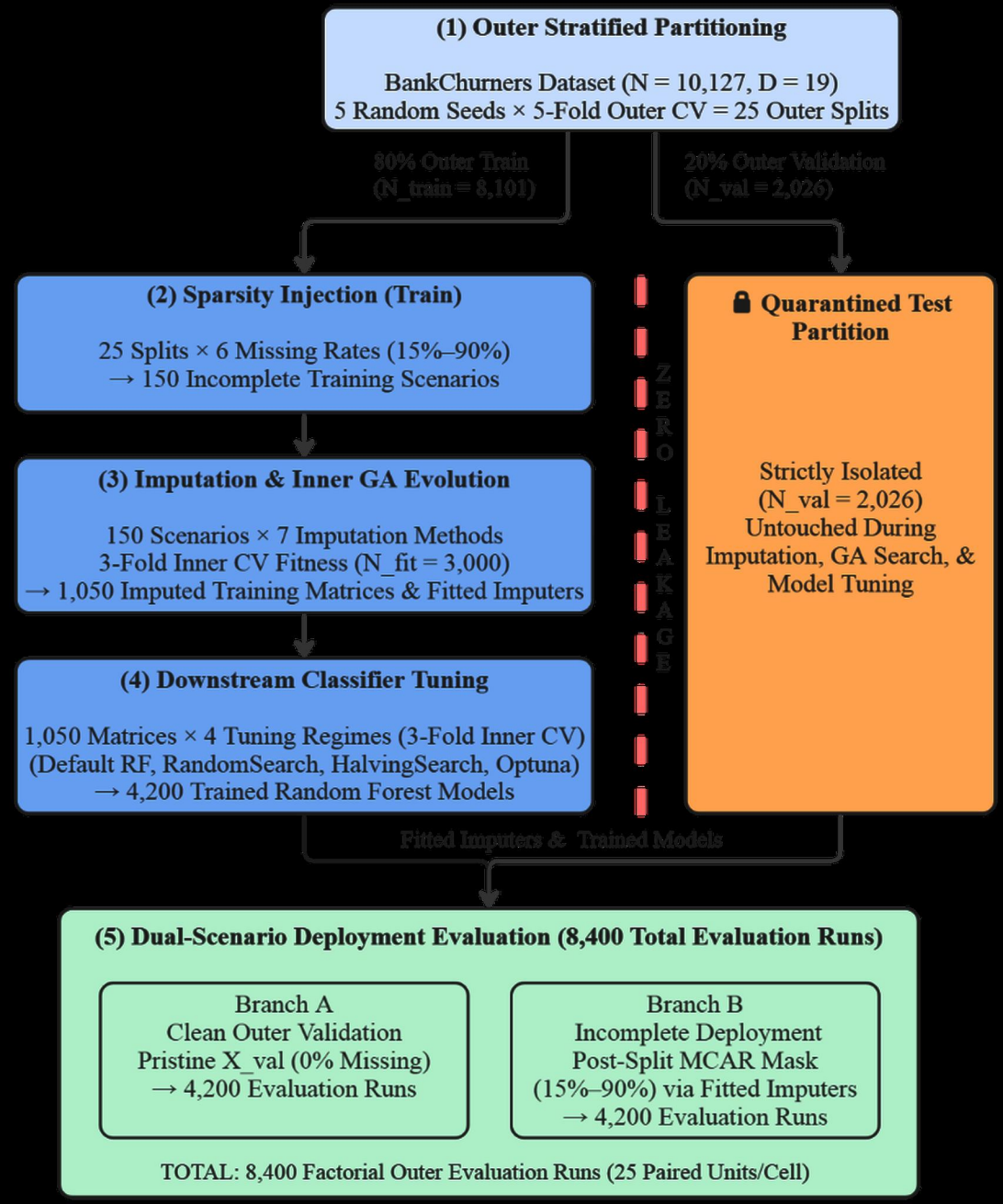

(1) Outer Stratified Partitioning
BankChurners Dataset (N = 10,127, D = 19)
5 Random Seeds × 5-Fold Outer CV = 25 Outer Splits
80% Outer Train
(N_train = 8,101)
20% Outer Validation
(N_val = 2,026)
(2) Sparsity Injection (Train)
25 Splits × 6 Missing Rates (15%–90%)
→ 150 Incomplete Training Scenarios
(3) Imputation & Inner GA Evolution
150 Scenarios × 7 Imputation Methods
3-Fold Inner CV Fitness (N_fit = 3,000)
→ 1,050 Imputed Training Matrices & Fitted Imputers
(4) Downstream Classifier Tuning
1,050 Matrices × 4 Tuning Regimes (3-Fold Inner CV)
(Default RF, RandomSearch, HalvingSearch, Optuna)
→ 4,200 Trained Random Forest Models
ZERO LEAKAGE
Quarantined Test Partition
Strictly Isolated
(N_val = 2,026)
Untouched During
Imputation, GA Search, &
Model Tuning
Fitted Imputers & Trained Models
(5) Dual-Scenario Deployment Evaluation (8,400 Total Evaluation Runs)
Branch A
Clean Outer Validation
Pristine X_val (0% Missing)
→ 4,200 Evaluation Runs
Branch B
Incomplete Deployment
Post-Split MCAR Mask
(15%–90%) via Fitted Imputers
→ 4,200 Evaluation Runs
TOTAL: 8,400 Factorial Outer Evaluation Runs (25 Paired Units/Cell)